Exam Date & Time: 31-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH BIOMEDICAL ENGINEERING END SEMESTER MAKE-UP EXAMINATIONS, DEC 2024

ANATOMY AND PHYSIOLOGY [BME 2121]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

1) Classify the epithelium. (3)

A)

B) Name the arteries supplying the heart and write a note on clinical implications / applied anatomy of arteries supplying heart. (3)

C) Describe the features of the lungs and mention the structure covering the lung. (4)

2) Mention the parts of the stomach and name the arteries supplying the stomach. (3)

A)

B) Mention the components of the urinary system and add a note on ureter. (3)

C) Name the endocrine glands and write a note on pituitary gland. (4)

3) Name the parts of the ear. (2)

A)

B) Mention the components of basal nuclei and write a note on clinical implications / applied anatomy of basal nuclei. (3)

C) Define synapse. Describe the sequence of events occurring during excitatory post synaptic potential. (5)

4) Describe the distribution of total body water. (3)

A)

B) Describe any TWO functions of blood. Mention ONE condition in which RBC count decreases. (3)

C) Draw and label the electrocardiogram (ECG) and mention the cause for each wave. (4)

5) Mention any TWO functions of middle ear and describe any ONE function. (3)

A)

B) Mention THREE uses of electromyogram (EMG). (3)

C) Name any TWO lung volumes and any TWO lung capacities and mention their normal values. (4)

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